RAHUL KAUSHAL

AI/ML Platform Engineer

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SUMMARY

I am a dedicated Software Engineer with a strong background in data engineering and machine learning operations. With my expertise in FastAPI, PySpark, and cloud technologies, I have successfully built cutting-edge platforms that empower data scientists and optimize model deployment. My proven track record in enhancing efficiency and reducing costs showcases my commitment to delivering high-quality solutions

EXPERIENCE

Software Engineer II

Honeywell

07/2024 | India

- Built MLOps platform using FastAPI enabling 15+ data scientists to train and deploy ML models to Databricks and Azure Kubernetes, supporting 35+ production models monthly with 99.5% deployment success rate
- Owned complete deployment system that automatically packages models into containers and deploys to Kubernetes or Databricks, reducing deployment time from 3 days to under 30 minutes and cutting infrastructure costs by 35% through smart container sharing
- Integrated 5 security scanning tools (Coverity, ModelScan, TwistLock, Malware, BDH) to automatically check uploaded code and models before deployment, catching 40+ critical vulnerabilities and meeting SOC 2 compliance standards
- **Tech Stack:** FastAPI, Python, Databricks, Azure Kubernetes Service (AKS), Docker, REST APIs, MLflow, Container Orchestration, Helm, Security Scanning (Coverity, ModelScan, TwistLock), CI/CD, Azure Cloud

Data Engineer II

Honeywell

07/2023 - 07/2024 | India

- Built and deployed large-scale data pipelines using PySpark and Databricks to process 500K+ service tickets and 10K+ PDF documents weekly, powering AI chatbot for 10+ enterprise customers
- Created automated PDF parsing system using PySpark and PyMuPDF to extract data from technical manuals and schematics, achieving 95% accuracy and eliminating manual data entry
- Developed real-time data ingestion system with change data capture (CDC), reducing data update time from 24 hours to under 15 minutes for critical service tickets
- Improved data pipeline efficiency by 70% through automated quality checks and schema validation, cutting ML team's development time from weeks to days
- **Tech Stack:** PySpark, Databricks, Apache Spark, PostgreSQL, Python, SQL, Delta Lake, PyMuPDF, unstructured.io

PROJECTS

SplitEase

11/2025

Expense Sharing & Budgeting App

- Building a Splitwise-like expense sharing and personal budgeting app using Flutter for the frontend and FastAPI with PostgreSQL for the backend, designed to support 100K+ users with features like group expenses, personal budgets, recurring payments (rent, insurance), and CSV import/export
- Implemented secure authentication with JWT access and refresh tokens, password hashing, and Redis-based rate limiting to protect REST APIs from abuse while maintaining low-latency responses
- Added automated testing for the backend using Behave for integration tests and Locust for performance testing, helping ensure the FastAPI and PostgreSQL stack scales reliably to 100K+ users

depkeeper

12/2025

Open Source Python Dependency Management Tool

- Developed Python library for pip users to manage dependencies, implementing conflict resolution, CVE security detection, and automated health scoring.
- Built high-performance async CLI tool with concurrent package analysis, supporting 5+ format conversions (requirements.txt, pyproject.toml, Pipfile, Poetry, Conda) and reproducible lock file generation using constraint satisfaction algorithms.
- Published to PyPI with comprehensive test coverage, CI/CD pipeline, and rich terminal UI using async/await patterns for maximum performance.

EDUCATION

M.Tech, Data Science

IIT Madras

05/2022 - 05/2023 | India

CGPA **8.77** / 10

B.Tech, Biological Engineering

IIT Madras

07/2018 - 05/2022 | India

GPA **8,77** / 10

KEY ACHIEVEMENTS

Enhanced Deployment Efficiency

Reduced deployment time by 90% from 3 days to under 30 minutes.

Streamlined Model Deployment

Achieved 99.5% success rate with 35+ production models monthly.

Cost Optimization Achieved

Cut infrastructure costs by 35% via smart container sharing.

Large-Scale Data Pipeline Built

Processed 500K+ tickets weekly for AI chatbot efficiency.

SKILLS

FastAPI | Python | REST APIs | CI/CD | Azure Databricks | PySpark | MLflow | TensorFlow | Keras | Kubernetes | Docker | Container Orchestration | Azure Cloud | SQL | PostgreSQL | Redis | GitHub Actions | Integration Testing | BDD | MLOps | Apache Spark | Flutter | Automated Testing

FIND ME ONLINE

LinkedIn
https://www.linkedin.com/in/rahul-kaushal04/

GitHub
https://github.com/rahulkaushal04

LANGUAGES

English
Proficient

Hindi
Native

Punjabi
Native